STRENGTH

Achilles and Knee Prehab Guide

The 15-minute routine attached to every run day, plus the load-management rules.

The run-day prehab routine (about 15 minutes)

Do this after every treadmill session, and on Monday with Strength A. Order matters: heel drops first while the calves are warm.

Exercise	How	Protects
Eccentric heel drops, straight knee (Alfredson)	Stand on a step on the balls of both feet. Rise up on both, shift to the target leg, lower slowly over 3 seconds until the heel is below the step. Use both legs to come back up. 3 x 15. Mild discomfort is acceptable, sharp pain is not.	Achilles tendon
Eccentric heel drops, bent knee	Same as above with the knee bent about 20 degrees. 3 x 15. Targets the soleus, the muscle that takes most of the load when running slowly.	Achilles / soleus
Seated or bent-knee calf raise	Knee at 90 degrees, weight on the thigh (dumbbell or plate), raise heel slowly. 3 x 15. Build to heavy over the year.	Soleus
Tibialis raise	Back against a wall, feet 30 cm out, lift toes toward shins. 3 x 20.	Shin / ankle
Single-leg balance	Stand on one foot 30-45 s, eyes open then closed. Progress to standing on a folded towel.	Ankle / knee control
Glute bridge	Lying on back, feet flat, drive hips up squeezing glutes, 2 s hold at top. 3 x 12. Progress to single leg.	Hips / knee
Clamshell with band	Side-lying, knees bent, band above knees, open the top knee without rolling the pelvis back. 3 x 15 each side.	Glute medius / knee tracking
Monster walk	Band above knees, quarter squat, step sideways 10 steps each direction x 3.	Glute medius
Step-down	Stand on a low step (10-15 cm). Slowly lower the other heel to the floor, tap, return. Keep the knee over the second toe. 3 x 10 each side.	Knee (patellofemoral)
Wall sit	Back to wall, thighs parallel-ish, 3 x 30-45 s. Progress duration, not depth.	Quads / knee
Terminal knee extension with band	Band around the back of the knee anchored in front, straighten the knee against the band. 3 x 15.	VMO / knee
Calf stretch (gastroc and soleus)	Straight-knee wall stretch 30 s, then bent-knee 30 s, each side. After sessions only.	Achilles / calf
Hip flexor stretch	Half-kneeling, tuck the pelvis, lean forward slightly, 30-45 s each side.	Hips / low back
Foam roll	Calves, quads, IT band region, glutes. 60 s each. Not on the Achilles tendon itself.	Recovery

Load-management rules for the run

- Treadmill only, 1% incline, cushioned deck. Shoes replaced every 500-600 km; note the start date on the sole.
- Run/walk from the first session. The walk breaks are the point: they let the tendon recover between loads.
- Rate pain 0-10 during and the morning after. 0-3 = continue as planned. 4-5 = repeat this week. 6+ = stop running for a week, keep swimming and cycling, do prehab daily, then restart one week back.
- Warm up the calves before every run: 5 min brisk walk, 20 slow calf raises, 10 leg swings each direction.
- Never run on consecutive days in Months 1-4.
- Body-weight check-in monthly. As weight drops, run volume can climb; the plan is re-written month by month with that in mind.